

00_summary_for_supervisor

March 30, 2026

1 Thesis Progress Summary

This notebook gives a short overview of the current thesis progress: - datasets used - preprocessing and multimodal merging - next possible steps

1.1 1. Datasets

The thesis uses: - PaySim for tabular fraud detection - Document image datasets for image-based fraud detection - A synthetic merged multimodal dataset for multimodal experiments

```
[1]: from pathlib import Path
import pandas as pd

project_root = Path.cwd().parent
processed_dir = project_root / "data" / "processed"

paysim_train = pd.read_csv(processed_dir / "paysim_train.csv")
paysim_val = pd.read_csv(processed_dir / "paysim_val.csv")
paysim_test = pd.read_csv(processed_dir / "paysim_test.csv")

image_train = pd.read_csv(processed_dir / "image_train.csv")
image_val = pd.read_csv(processed_dir / "image_val.csv")
image_test = pd.read_csv(processed_dir / "image_test.csv")

mm_train = pd.read_csv(processed_dir / "mm_train.csv")
mm_val = pd.read_csv(processed_dir / "mm_val.csv")
mm_test = pd.read_csv(processed_dir / "mm_test.csv")

print("PaySim:")
print(paysim_train.shape, paysim_val.shape, paysim_test.shape)

print("\nImage:")
print(image_train.shape, image_val.shape, image_test.shape)

print("\nMultimodal:")
print(mm_train.shape, mm_val.shape, mm_test.shape)
```

PaySim:

(4453834, 12) (636262, 12) (1272524, 12)

Image:
(6653, 10) (1743, 10) (2425, 10)

Multimodal:
(6653, 29) (1658, 29) (2425, 29)

```
[4]: summary = pd.DataFrame({
    "Dataset": ["PaySim", "Image", "Multimodal"],
    "Train": [len(paysim_train), len(image_train), len(mm_train)],
    "Validation": [len(paysim_val), len(image_val), len(mm_val)],
    "Test": [len(paysim_test), len(image_test), len(mm_test)]
})

summary
```

```
[4]:
```

	Dataset	Train	Validation	Test
0	PaySim	4453834	636262	1272524
1	Image	6653	1743	2425
2	Multimodal	6653	1658	2425

1.2 2. Label distributions

```
[ ]: def show_distribution(name, df, target_col):
    print(f"\n{name}")
    print(df[target_col].value_counts())
    print(df[target_col].value_counts(normalize=True) * 100)

show_distribution("PaySim train", paysim_train, "isFraud")
show_distribution("Image train", image_train, "original_label")
show_distribution("Multimodal train", mm_train, "final_label")
```

```
PaySim train
isFraud
0    4448085
1      5749
Name: count, dtype: int64
isFraud
0    99.87092
1     0.12908
Name: proportion, dtype: float64
```

```
Image train
original_label
0           2796
forged       2012
attack       1230
```

```

bonafide      615
Name: count, dtype: int64
original_label
0             42.026154
forged        30.241996
attack        18.487900
bonafide       9.243950
Name: proportion, dtype: float64

```

```

Multimodal train
final_label
0      3411
1      3242
Name: count, dtype: int64
final_label
0      51.270104
1      48.729896
Name: proportion, dtype: float64

```

1.3 3. Multimodal merge logic

I first analyzed the tabular and image datasets separately. Then I merged the image datasets into one unified image dataset with binary labels. Finally, I merged this image dataset with PaySim using a synthetic pairing strategy to create the multimodal dataset.

Since the tabular and image datasets do not have natural one-to-one matching samples.

To create the multimodal dataset, I used the following rules: - samples were matched only within the same split (train, validation, test) - samples were matched only within the same binary label - samples were shuffled before pairing - the final number of multimodal pairs in each split and class was limited by the smaller modality

This created a synthetic but label-aligned multimodal dataset without split leakage.

```

[3]: cols_to_show = [
    "mm_id",
    "split",
    "final_label",
    "tab_step",
    "tab_type",
    "tab_amount",
    "img_image_path",
    "img_image_class",
    "img_source_dataset"
]

mm_train[cols_to_show].head(10)

```

```
[3]:
```

	mm_id	split	final_label	tab_step	tab_type	tab_amount	\
0	mm_train_1_002875	train	1	343	4	802417.45	
1	mm_train_0_000217	train	0	251	3	40693.51	
2	mm_train_0_003214	train	0	188	1	279091.56	
3	mm_train_1_001104	train	1	17	1	1639676.27	
4	mm_train_0_001544	train	0	252	0	79797.27	
5	mm_train_0_000247	train	0	349	1	313269.65	
6	mm_train_1_002906	train	1	702	4	10000000.00	
7	mm_train_0_000230	train	0	404	3	442.49	
8	mm_train_1_001470	train	1	215	1	319887.08	
9	mm_train_1_002379	train	1	726	1	1539474.57	

	img_image_path	img_image_class	\
0	C:\Users\Admin\Desktop\thesis\multimodal-fraud...	forged	
1	C:\Users\Admin\Desktop\thesis\multimodal-fraud...	bona_fide	
2	C:\Users\Admin\Desktop\thesis\multimodal-fraud...	bona_fide	
3	C:\Users\Admin\Desktop\thesis\multimodal-fraud...	forged	
4	C:\Users\Admin\Desktop\thesis\multimodal-fraud...	bona_fide	
5	C:\Users\Admin\Desktop\thesis\multimodal-fraud...	bona_fide	
6	C:\Users\Admin\Desktop\thesis\multimodal-fraud...	forged	
7	C:\Users\Admin\Desktop\thesis\multimodal-fraud...	bona_fide	
8	C:\Users\Admin\Desktop\thesis\multimodal-fraud...	forged	
9	C:\Users\Admin\Desktop\thesis\multimodal-fraud...	forged	

	img_source_dataset
0	fantasyid
1	fantasyid
2	midv
3	fmidv
4	midv
5	midv
6	fmidv
7	midv
8	fantasyid
9	fantasyid

1.4 4. Experiment setup

Current planned experiments: 1. Tabular-only model on full PaySim 2. Tabular-only model on merged subset 3. Image-only model on full image dataset 4. Image-only model on merged subset 5. Multimodal model on merged subset

1.5 5. Current observations

- The multimodal dataset was created successfully.
- The merged dataset contains both tabular PaySim features and image paths.
- The multimodal dataset is smaller than the original datasets because pairing is limited by the smaller class size in each split.

- A fair comparison is possible on the merged subset.

1.6 6. Next steps

The next step is to train and compare: - full tabular baseline - merged tabular baseline - full image baseline - merged image baseline - merged multimodal baseline